

Lab 18: Evolution — Tragedy of the Commons and Game Theory (Fishing Simulation)

BIOL-8

Name: _____ Date: _____

Group:

Purpose

You will work in a **group** to fish a **shared stock** of four kinds of fish over **three seasons**.

The activity connects **evolution and ecology** to **human behavior** in two ways:

- **Tragedy of the commons** — a shared resource can be depleted when many people pursue short-term private gain.
- **Game theory** — your payoff depends on others' choices; **repeated** seasons change what looks like a good strategy.

After each season, fish that were **not** caught can **reproduce** using a fixed rule. The pool can **recover** or **collapse**, depending on what the group harvests.

Learning objectives

By the end of this lab, you will be able to:

- **Explain** the tragedy of the commons in your own words and relate it to a **shared** fish pool your group managed.
- **Describe** how **rotating** who fishes each season and **rising point values** (each season pays more per fish) can change incentives from one round to the next.
- **Apply** the reproduction rule to update population sizes **per species** after a season.
- **Connect** the simulation to **game theory** terms where appropriate: strategy, payoff, cooperation vs. short-term gain, and effects of **repeated** interaction across seasons.
- **Compare** the **four species** by value: gold vs. silver and large vs. small.

Background

Tragedy of the commons

When a **resource** (pasture, fishery, clean air) is **open to many users** and no one person pays the full **future** cost of overuse, each user can be tempted to take as much as possible **now**. If everyone does that, the **shared stock** shrinks and may **collapse**—even though **no one** wanted that outcome. **Rules, quotas, catch limits, and community agreements** exist partly to avoid that outcome. This lab is a **toy** version of that tension.

Game theory in one paragraph

Game theory models choices when your outcome depends on **others'** choices. In a **repeated** setting (here, **three seasons**), what pays off in season 1 can differ from what pays off after others respond in seasons 2 and 3. You may see **defection** (take a lot) vs. **cooperation** (leave fish for the group and the future) or a mix, depending on your group's **norms** and the **rising point values** in the schedule.

The four species and value (relative prices)

Relative value (highest to lowest in this model):

Species	Value in the model
Large goldfish	Highest value per fish (gold and large)
Small goldfish	High, but less than large gold (still gold)
Large silverfish	Less than either gold (silver), but large > small for silver
Small silverfish	Lowest value per fish

Gold types are worth more than **silver** types. **Large** types are worth more than **small** types, within the same “metal” group.

Prices and seasons: Earnings are in **class points** per fish, using the **fixed schedule** in the **Data** section below (**Table 2**), unless your instructor posts a different table. **Every** species earns **more points** in **season 2** than in **season 1**, and more again in **season 3**—so the pull to overharvest can grow each round if your group does not protect the stock.

Reproduction rule (after each season, before the next)

Work **separately for each of the four species**:

1. Count how many fish of that species were **not** caught (the fish **left in the pool** after that season's harvest). Call this **N** (a whole number).

2. For that species, the pool **gains** new fish. **Divide N by 2 and round down** to a whole number (in math this is the *floor* of $N/2$). That number is how many **new** fish of that species you **add**.
3. Example: $N = 5 \rightarrow 5 / 2 = 2.5 \rightarrow 2$ new fish (round down). New total = $5 + 2 = 7$ before the next season.
4. The **caught** fish are **removed** from the pool. They do **not** count toward reproduction in this model.

If your instructor uses a **different** reproduction rule, **follow their version**.

Connection to “evolution” in BIOL-8

Evolution is not only long-term **genetic** change. Biologists also study how **ecological and social** conditions **select** for behaviors, norms, and structures (e.g. cooperation, punishment of cheaters) that affect who survives and **reproduces**—including in **H. sapiens**. This lab is a **classroom analogy**: no real evolution in the fish, but real discussion about **incentives, group outcomes**, and **sustainability**—concepts that sit next to **natural selection** in a broader life-science view.

Materials

- **Shared pool** of countable units for four species (e.g. colored chips, small cards, or tallies) — **large gold, small gold, large silver, small silver**
 - A way to record **start, catch, left, reproduction add**, and **end counts per species, per season**
 - The **point schedule** printed in this handout (Table 2) — *your instructor may substitute; use their values if so*
 - This handout and writing tools
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Procedure

Before season 1

1. **Form groups** and write **all members’ names** in the roster below. You will use **three fishing seasons**.
2. For **each season, one** group member is the **fisher** for that season (someone who was not the fisher the previous time). **Rotate** so **three different people** fish in seasons 1–3 (if the group has fewer than three people, your instructor will say how to handle rotation).

3. Copy the **starting population** for each **species** from the instructor (or the board) into **Table 1** under **Start (Season 1)**.
4. Use the **price schedule (points per fish)** printed in **Table 2** below for all earnings, **unless** your instructor gives a different table—then use theirs for every season.

Each season (repeat for seasons 1, 2, and 3)

1. The **fisher** for that season “catches” by **taking** units from the pool as allowed by the rules your instructor gives (e.g. **maximum** number of fish, or a **time** limit, or a **fixed** haul). Record **only** the **number caught per species** in **Table 1** in the **Catch** row for that season.
2. **After** the catch, compute **left** in the pool: **Start – catch** (per species) for that season. Enter under **Left after catch**.
3. **Reproduction:** For **each** species, compute **add** = **(Left after catch) / 2, rounded down** to a whole number. New count before the next season = **Left after catch + add**. Enter under **Repro add** and **End (after repro)**. The **End** row is the **Start** for the next season.
4. **Earnings for this season:** For each species, **(catch) × (points for that species in this season)** from the **Table 2** price grid. Add the four products to get **group points** for the season. Do the same for seasons 2 and 3 using the **points columns** for that season in the same table.
5. The **next** group member **fishes** the following season, starting from the new **Start** row.

After season 3

- Total **cumulative** earnings if your instructor requests it.
 - Complete **Table 1** and **Table 2**, then the **Analysis** and **Conclusion** sections.
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Data

Abbreviations: LG = large gold, SG = small gold, LS = large silver, SS = small silver.

Show at least one season's earnings as a sum of (catch $\times$ points) for each species (in Table 2 or the work space below the price table).

Group roster (fill in before Season 1)

Group roster

Role	Name
Fisher — Season 1	
Fisher — Season 2	
Fisher — Season 3	
Other group members (if any)	

Table 1 — Population and catch (all three seasons)

Starting populations (from instructor) go in the **Start — Season 1** row below.

Table 1 — Season 1

	Large gold (LG)	Small gold (SG)	Large silver
Start — Season 1			
Catch — Season 1			
Left after catch — S₁			
Repro add — S₁ (left / 2, round down)			
End after repro — S₁ (= left + add)			

Season 2 — *Fisher (name):*

Table 1 — Season 2

	LG	SG	LS
Start (= end S1)			
Catch			
Left after catch			
Repro add (left / 2, round down)			
End after repro			

Season 3 — *Fisher (name):*

Table 1 — Season 3

	LG	SG	LS
Start (= end S2)			
Catch			
Left after catch			
Repro add (left / 2, round down)			
End after repro			

Table 2 — Price schedule (class default) and earnings

Points per fish caught — use these values for **earnings** unless your instructor replaces them.

Gold rows earn more than **silver**; **large** more than **small**; each **season** pays more per fish than the last.

Species	Points — Season 1	Points — Season 2	Points — Season 3
LG (large gold)	8	12	18
SG (small gold)	4	6	9
LS (large silver)	2	3	5
SS (small silver)	1	2	3

Earnings for a season = **sum** over the four species of **(number caught) × (points for that species in that season)**.

Work space (one season, optional check): Pick **one** season and write **(catch × points)** for each species, then the **sum** (should match the row you enter in the Earnings table below).

Table 2 — Earnings by season (points)

Season	Group earnings (show work or subtotal)
1	
2	
3	
Total (all seasons)	

Analysis

Answer in **complete sentences**. Use your **tables** for any question that refers to your group's numbers.

1. Tragedy of the commons. In two or three sentences, did your group's pool **trend toward depletion, stability, or recovery** for one or more species? Name **one** choice that made that outcome more likely.

2. Evolution of incentives. Prices rose each season. How did that affect what you think the "best" **short-term** move was, versus a **long-term** group goal?

3. Game theory. Name **one** behavior that acted like **cooperation** and **one** that acted like **defection** (or self-interest only), whether or not you used those words during the game.

4. Reproduction rule. For **one** species, show with numbers: **left after catch, repro add** (left / 2, **round down**), and **end** after you add. Did the rule ever **increase** the stock enough that your group was glad some fish were **left in the pool**?

Conclusion

5. Human biology in one sentence: Why might a **lecture** on evolution in humans include a **resource** and **incentive** story like this, not only DNA and fossils?

Lab 18 — Evolution. Tragedy of the commons and game theory via a three-season fishing simulation.